

Household Energy Intelligence

Detailed Project Report | Nitin Kohli

Internship-style Data Analytics + Data Science portfolio project

Project focus: household electricity demand, forecasting, usage-pattern clustering and practical sustainability analytics.

Verified project snapshot

Metric	Result
Raw records	1,048,575
Rows with missing	4,069 (0.39%)
Usable days	725
Usable hours	17,402
Best model	Gradient Boosting
Best R2	0.66
MAE reduction	40% vs weekly baseline
High-use RF accuracy	80%
Clusters	3

1. What I built

I treated the supplied CSV like an internship project rather than a single notebook exercise. The workflow starts with data quality, creates clean daily and hourly tables, explains usage patterns, compares practical machine-learning models, benchmarks neural models, groups daily demand shapes, and finishes with SQL, Excel and Power BI-ready outputs.

Core question: How can historical household electricity data help explain demand, estimate near-term usage and identify patterns that are useful for better energy planning?

2. Data and quality checks

Check	Finding
Records	1,048,575
Original fields	8 measurement columns + date/time
Period	2006-12-16 to 2008-12-13
Missing rows	4,069 (0.39%)
Duplicate rows	0
Duplicate timestamps	0
1-minute interval consistency	100.0%
Longest missing streak	62.0 hours

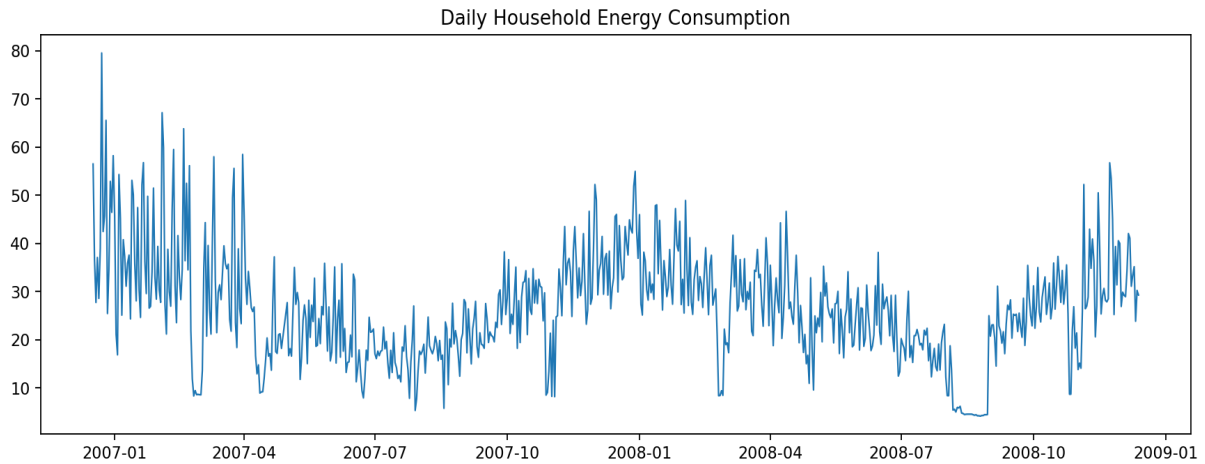
Cleaning choice: short gaps were interpolated only for small-window analysis. Long gaps were not blindly filled. Daily and hourly modeling kept time buckets with at least 90% coverage.

3. EDA and statistical analysis

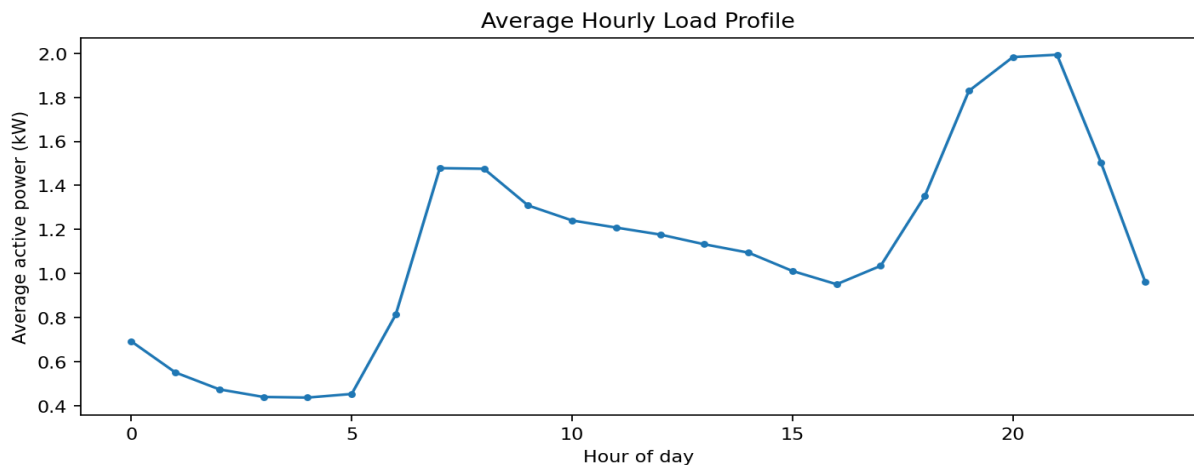
Average daily energy is 26.57 kWh/day. Weekday use is 25.01 kWh/day and weekend use is 30.52 kWh/day, which is about 22% higher on weekends.

The strongest average demand period is the evening, especially around 19:00-21:00. This is useful for peak-demand discussions because it gives a clear time window to watch.

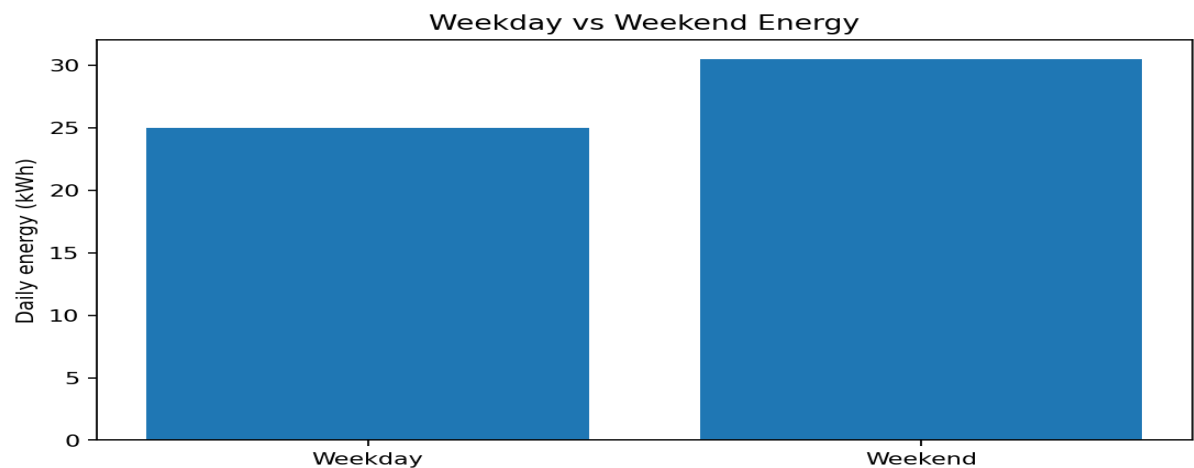
Daily energy trend



Average hourly load profile



Weekday vs weekend



4. Feature engineering

The forecasting table uses simple features that are easy to explain and reproduce:

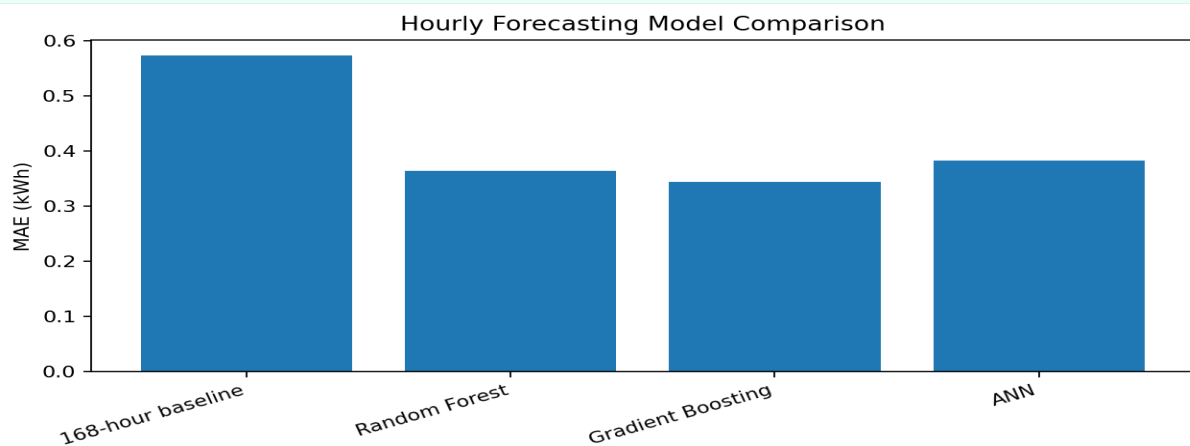
1. Hour of day
2. Day of week
3. Month
4. Weekend flag
5. Lag 1 hour
6. Lag 24 hours
7. Lag 168 hours (same hour last week)
8. 24-hour rolling mean using past data
9. 168-hour rolling mean using past data

I used lags because energy consumption is time dependent. Lag 24 captures the previous day at the same hour, while lag 168 captures the previous week at the same hour.

5. Machine-learning comparison

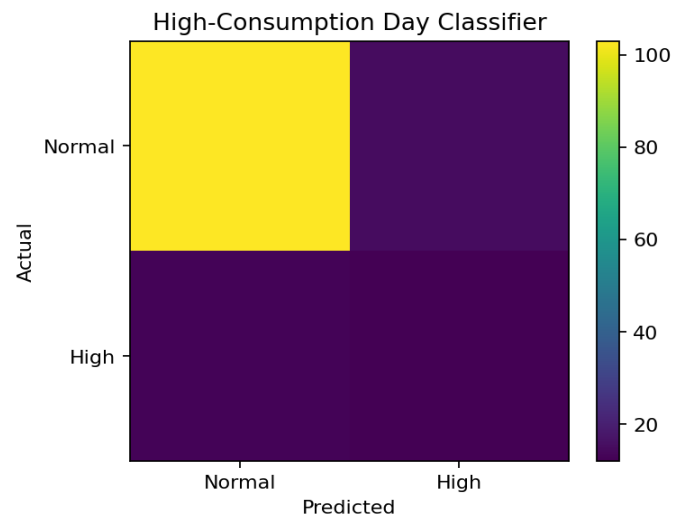
Model	MAE	RMSE	R2
168-hour baseline	0.572	0.897	-0.004
Random Forest	0.364	0.520	0.663
Gradient Boosting	0.344	0.525	0.657
ANN	0.382	0.549	0.625

Final practical choice: Gradient Boosting. It reached $R^2 = 0.657$ with $MAE = 0.344$ kWh and reduced MAE by 40% versus the 168-hour baseline.



6. High-consumption classification

For an alert-style problem, a day above the 75th percentile of training-set energy was labelled high-consumption. Random Forest reached 80% accuracy on the time-based test set, with 48% recall for the high-use class.



Important interview point: 80% accuracy belongs to the classification task. It is not the forecasting accuracy of the Gradient Boosting model.

7. ANN and LSTM benchmark

LSTM test results were MAE 0.411 kWh, RMSE 0.578 kWh and R2 0.58.

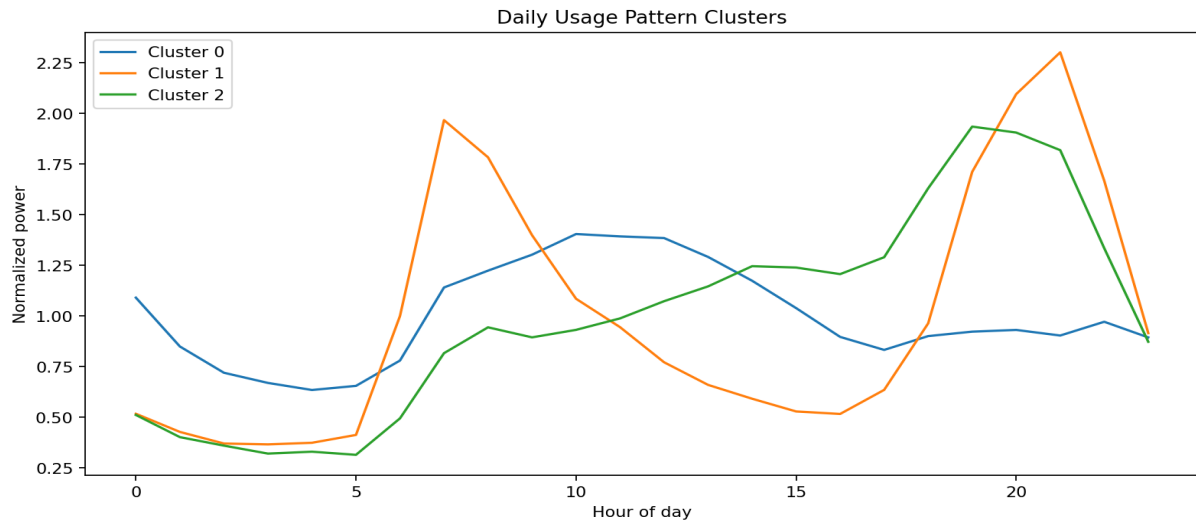
The neural models are benchmarks; the practical final model is Gradient Boosting.

learning was tested, but the simpler Gradient Boosting model performed better for the final business use case.

Why no CNN? The dataset is tabular and time-series based, not image or spatial-grid data. I did not force CNN into the project just to add another algorithm name. This is a good example of choosing a method because it fits the data.

8. K-Means clustering

I converted each usable day into a 24-hour load profile, normalized the profile and grouped 723 daily profiles into 3 usage-shape clusters.



9. SQL, Excel and Power BI

SQL: business queries for averages, top-consumption days, hourly patterns, model comparison and rolling averages using a window function. Excel: KPI summary, daily energy, hourly profile, model results and cluster profiles. Power BI: three-page plan - Energy Overview, Forecasting, and Sustainability Patterns. The project includes ready-to-import CSV tables and a dashboard guide.

10. Sustainability interpretation

The strongest sustainability signal is demand timing. The sample shows a clear evening peak and a 22% weekend uplift versus weekdays. These patterns can support practical actions such as shifting flexible loads, monitoring peak windows and comparing routines over time.

I did not claim a specific CO2 or cost saving percentage because the dataset does not contain a verified emissions factor, electricity tariff or household context. That keeps the sustainability story honest.

11. Four resume bullets

1. Analyzed 1,048,575+ minute-level household energy readings using Python, Pandas, NumPy, EDA and statistical analysis, cleaning 0.39% affected records and building daily/hourly datasets for business analysis.
2. Built an hourly time-series forecasting pipeline with feature engineering, lag/rolling features, Random Forest, Gradient Boosting and ANN, reaching R2 0.66 and reducing MAE by 40% versus a weekly-history baseline.
3. Benchmarked ANN and LSTM sequence models and added a high-consumption classification workflow, with Random Forest reaching 80% accuracy on the time-based test period for energy-use alerts.
4. Used K-Means clustering, SQL, Excel and Power BI-ready dashboards to segment 723 daily load profiles into 3 usage patterns, highlight a 22% weekend uplift, and convert energy trends into practical sustainability actions.

12. What each resume number means

Resume number	Meaning
1.05M+	Total rows in the supplied CSV. It shows data volume.
0.39%	Rows affected by missing numeric readings: 4,069 of 1,048,575.
R2 0.66	The selected hourly forecast model explains about 66% of test-period target variance.
40%	MAE reduction compared with the 168-hour weekly-history baseline. It does not mean 40% accuracy.
80%	Test accuracy for the high-consumption-day Random Forest classifier.
3 clusters	Three daily usage-shape groups created with K-Means.
22%	Weekend average daily energy is around 22% above weekday average in this sample.

13. HR and interview questions

Tell me about the project in 60 seconds.

I worked on a household electricity dataset with over one million minute-level records. I cleaned the data, created daily and hourly views, performed EDA and statistics, then built time-aware forecasting, high-use alerts and K-Means clusters. I compared tree and neural models and selected Gradient Boosting because it gave the strongest practical test results.

Why did you aggregate minute data?

The raw data is very large and noisy. Hourly and daily views are easier to explain, model and use in dashboards while keeping the key demand patterns.

How did you handle missing data?

I did not fill every gap. I interpolated only short gaps for small-window analysis and used a 90% coverage rule for daily/hourly modeling. A roughly 62-hour missing streak was left out instead of inventing values.

Why a time-based split?

Forecasting should learn from the past and be tested on later observations. Random splitting can leak future patterns into training.

Why Gradient Boosting?

It produced the lowest MAE and a strong R2 among the practical forecast models tested.

What does 40% improvement mean?

The forecast MAE is about 40% lower than the weekly-history baseline. It is an error reduction, not 40% accuracy.

Why lag 24 and lag 168?

24 hours captures the previous day at the same hour; 168 hours captures the same hour one week earlier.

Why ANN and LSTM?

ANN is a reasonable neural model for the tabular features. LSTM is a natural sequence benchmark for hourly demand.

Why not CNN?

There is no image or spatial-grid structure. CNN would add complexity without a clear reason.

Why K-Means?

It groups similar 24-hour daily load shapes and is easy to interpret for segmentation.

How did SQL help?

I used SQL to answer business questions on the processed daily/hourly tables, including averages, top days and rolling calculations.

How did Power BI help?

It turns the processed tables into an interactive view for non-technical users. I planned overview, forecasting and sustainability pages.

What is the biggest limitation?

The data shows what happened, but not all the reasons. Weather, occupancy, tariff and appliance context are missing, so I avoid claiming causation.

What would you improve next?

Add weather, tariff and occupancy data, then quantify cost and carbon impact with verified factors. Only add more complex models when they improve the business metric.

14. GitHub walkthrough

The repository is intentionally limited to five main folders so a recruiter can understand it quickly:

- 01_Data - raw input, data dictionary, processed daily/hourly tables and model outputs
- 02_Notebooks - six Jupyter notebooks in project order
- 03_Analytics_BI_SQL - SQL database/queries, Excel layout and Power BI guide
- 04_Visuals - EDA, ML, forecasting, clustering and dashboard images
- 05_Documentation - detailed report, resume bullets and verified project metrics

15. Limitations and honest scope

- The file ends in December 2008, so forecasts describe this historical sample.
- Weather, tariffs, occupancy and appliance metadata are not available.
- The high-consumption threshold is defined for this project; it is not a utility-company standard.
- Deep-learning results depend on sequence length, scaling and training settings.

Final project pitch

Built an end-to-end household energy analytics project that turned 1M+ minute-level readings into clean datasets, demand forecasts, high-use alerts, usage clusters and Power BI-ready sustainability insights.